4.4 Circulation

i Know the structure of the heart, arteries, veins and capillaries.

ii Understand the advantages of a double circulatory system in mammals over the single circulatory systems in bony fish, including the facility for blood to be pumped to the body at higher pressure and the splitting of oxygenated and deoxygenated blood.

iii Know the sequence of events of the cardiac cycle.

iv Understand myogenic stimulation of the heart, including the roles of the sinoatrial node (SAN), atrioventricular node (AVN) and bundle of His.

v Be able to interpret data showing ECG traces and pressure changes during the cardiac cycle.

vi Know the structure of blood as plasma and blood cells, to include erythrocytes and leucocytes (neutrophils, eosinophils, monocytes and lymphocytes).

vii Know the function of blood as transport, defence, and formation of lymph and tissue fluid.

viii Understand the role of platelets and plasma proteins in the sequence of events leading to blood clotting, including:

- platelets form a plug and release clotting factors, including thromboplastin
- prothrombin changes to its active form, thrombin
- soluble fibrinogen forms insoluble fibrin to cover the wound.

ix Understand the stages that lead to atherosclerosis, its effect on health and the factors that increase the risk of its development.